SOME OBSERVATIONS ON THE ATTITUDE OF WEST-EUROPEAN GOVERNMENTS TO THE DEVELOPMENT OF DEFENSIVE WEAPONS IN OUTER SPACE

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I. The West-European reaction to the fundamental principle of S.D.I.

The first question to be considered is whether, and if so to what extent, the European attitude is in conformity with that of the U.S. on this issue.

a) In answering this question, one is faced with the widely divergent views expressed in the U.S. on the needs, the feasibility and the cost-effectiveness of the S.D.I. programme. Whereas an enquiry in these divergencies would far exceed the scope of the present article, it is important to note that, notwithstanding the doubts expressed by several leading American scientists on the technological feasibility of the programme and notwithstanding the opposition of, in particular, several Democratic members of Congress to - at least the development of the *kind* of defensive weapons in space which the Administration wishes to pursue (see below) - on the basis of recent statements made by President Reagan, it can be concluded that he is determined to vigorously pursue the programme. It further should be noted that this determination has not been weakened in any sense by the continuing strong opposition of the Soviet Union to S.D.I. One of the main reasons why this opposition has been repudiated, can be sought in the American conviction that the Soviet Union, for its part, has since initiated a substantial research programme into advanced technologies for defense against ballisitic missiles. In this context reference may be made to a recently published "Guide to the S.D.I. Inititative," in which it was submitted that the Soviet defense programme covered many of the same technologies involved in S.D.I., but that this programme represented a far greater investment of plant space, capital and man power.[1]

In view of this conviction it is difficult to imagine that the new proposal presented by the Soviet Union to the U.S. in Geneva on 29 July 1987, in which the reduction of strategic nuclear arms is linked to strict limits on anti-missile systems, would prove to be acceptable to the American Administration. Soviet suggestions regarding such a link have been made before and have been constantly rejected by the U.S.

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1. *See* GUIDE TO THE SDI INITIATIVE, appendix A, (Pasha Publications, 1986.)

b) When investigating the West-European reaction to S.D.I. *in general*, attention should be drawn first to the conflicting views manifested in the European press.

As to Britain, the Editor of the "Sunday Telegraph", *Mr. Peregrine Worsthorne*, in a recent article, wrote that most West Europeans opposed S.D.I. The same view was expressed in an article of the "Economist", in which it was contended that public opinion in Western-Europe was anti-S.D.I., though it was suggested that this public was less likely to make a fuss about it, once Europeans would have the Euro-missile deal.[2]

Insofar as the West-German press is concerned, attention may be drawn to an article by the diplomatic correspondent of "Die Zeit", in which he observed that the West-European reaction to S.D.I. had wavered between opportunism and pusillanimity.[3]

On the basis of these statements, one might perhaps be inclined to conclude a general opposition in Western-Europe to the programme. But if one looks at the official attitude of West-European Governments, such a conclusion cannot be sustained.

In the first place, in October 1986, fourteen of the sixteen members of NATO, adopted a highly significant resolution supporting S.D.I. Only Greece and Denmark did not go along with this support.

Among the countries which originally took a negative position regarding the programme but later changed its mind, mention should be made of France. The Prime Minister, *Mr. Chirac*, recently declared that the French Government was strongly in favour of the development of S.D.I.

Secondly, the governments of Britain, West-Germany, France and Italy, all have given permission to companies in their countries to cooperate with the U.S. in the research of this programme.

Before proceeding to an investigation of the West European views on the *legal* aspects of S.D.I., I will refer to the following interesting comment on S.D.I., made by the former British Ambassodor in Washington, *Sir Oliver Wright*, in the Annual Memorial Lecture to the David Davies Memorial Institute of International Studies, entitled "Anglo-American Relations: The Atlantic Grows Wider":

> "Thank goodness we have not opted out of the S.D.I. programme; whether or not we shall create a celestial astrodome to protect us from the bomb, the technologies involved in finding out whether we can, will certainly affect the lives of successive generations, just as the microchip of the moonshot influences our lives today"[4]

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2. *See* The Economist, 18 October 1986.

3. *See* The International Herald Tribune, 14 November 1986, p. 1.

4. *See* Lecture by Sir Oliver Wright, Annual Memorial Lecture, David Davies Memorial Institute of International Studies 4 (Dec. 9, 1986).

II. The West European views on the kind of defensive weapons in space which are legally permissible

a) Again on the issue, some observations should be made first on the continuing conflicts between the American Adminstration and Congress on this matter.

The most crucial issue which - in this context - has arisen, concerns the interpretation of the ambigous provision laid down in Agreed Statement D, annexed to the ABM Treaty of 1972, which provides that, in the event of ABM systems based on other physical principles, are created in the future, limitations on such systems would be subject to discussions. Can this provision be interpreted in the sense it allows the *testing* of exotic weapons in space?

Although *President Reagan*, until 1985, took the position that, on the basis of this stipulation, advanced testing of these kinds of weapons would be allowed, he declared that the Government would not conduct such testing.

However, in the beginning of February 1986, *Mr. Schultz* stated that S.D.I. had yielded considerable progress even under the narrow interpretation of the Treaty that the Administration embraced, but that it was clear now, giving the progress that had been made, that one would be able to pursue this progress much more effectively - and perhaps only - if a different pattern of testing was permitted.

The most important and recent challenge to the broad interpretation of the Treaty has come from *Senator Sam Nunn*, Chairman of the Senate Armed Services Committee. In the week of March 12 1987, he submited to the Senate an analysis in which he stressed that after weeks of examining the classified negotiating record of the Treaty, of the open ratification debates and subsequent practice, he had not uncovered a single statement confirming that testing and development of space-based exotic technologies is allowed.

Though *Senator Nunn* has not yet declared where the ABM debate should go next, he predicted a serious constitutional confrontation if the Administration persisted in rewriting history with its implied disrespect for Congress in its Treaty-ratifying role.[5] It should be noted, however, that the Senator has supported the President's S.D.I.

As to the latest developments of the American S.D.I. research programme, it was reported in July 1987 that the S.D.I. managers had shifted the programme followed up till now, in favour of a crash programme for early development of a relatively unsophisticated minimal missile defense. The near-term deployment of this system, which the managers have in mind for 1994-95, would have no lasers or beam weapons

5. *See* Kevin L. Kennedy, *Treaty Interpretation by the Executive Branch: The ABM Treaty and Star Wars Testing and Deployment*, AM. J. INT. L. 854 (October 1986).

but only a token deployment of space-based kinetic kill vehicles. It has been suggested that such a crash effort was aimed at a "fait accompli", that would narrow the options of future Administrations, likely to have more knowledge of scientific prospects and Soviet responses.

At the time of writing the present article, no information has as yet become available on the reaction of the Administration to this crash programme. However that may be, as was mentioned above, the Administration appears to press forward with S.D.I. testing under the broad interpretation of the ABM Treaty and on 22 July 1987, a senior U.S. official declared that President Reagan had rejected suggestions by the Soviet Union and by arms control adviser, *Mr. Paul Nitze,* to negotiate what tests could be permitted under the Treaty.[6]

Attention should also be drawn to an article published in the New York Times on 21 August 1987 by *Mr. Weinberger* in which he expressed the view that with adequate funding for S.D.I. research "we would confidently anticipate that phased deployment could begin as early as 1994 or 1995."

In view of these latest developments, it seems highly unlikely that in the short remaining time before the election at the end of 1988, the present confrontation between the Administration and the majority of the Democratic members of Congress can be solved. It can be assumed that it will be the new Administration which will have to try and arrive at a solution of this controversial issue.

Though great caution is required if one wishes to try and speculate on the possible reaction to the development of the S.D.I. programme of a new Administration, there have been some recent statements which show that the Democratic members of Congress, as yet, have not reached a consensus on the way the S.D.I. programme should be pursued.

According to a recent article in the "Washington Post", an analysis of the views expressed by the Presidential candidates of the Democratic Party shows that all of them, though being inclined to be in favour of a ban on the testing of Defensive space weapons outside the laboratory, wish to continue with research on these weapons. However, in an article published in the British journal "The Economist" on 1 August 1987, reference was made to a declaration of *Senator Paul Simon* (who entered the Presidential race in May) according to which he would be inclined to do away with S.D.I.

But as a majority of the Democratic members of Congress appear to share the conviction of the present Administration that a continuation of the research on space weapons is essential in view of the substantial Soviet research programme of these weapons, can one really expect that a Democratic Administration would be prepared to abandon the S.D.I. programme altogether?

b) When one now assesses the attitude of the West European countries on the interpretation of the provisions of the Treaty, reference

6. *See* The International Herald Tribune, 19 June 1987, p. 1.

should be made first to a letter which *Lord Carrington*, Secretary General of NATO, sent to the American Administration in which NATO's great concern of the implications of some form of deployment of S.D.I. was expressed, and in which he asked for the Alliance to be consulted before a decision on this issue was made.

Britain, West Germany, France, the Netherlands and Belgium all let it be known that they considered the present Administration's broad interpretation to be legally unjustified and, moreover, that the testing of new space defensive weapons would have a serious political effect.

In this context, some remarks may be made on the criticism expressed by *Mr. Kenneth Adelman*, Head of the U.S. Arms Control and Disarmament Agency, on the European rejection of a broad interpretation of the Treaty. During a conference of American Military Veterans, held at the end of January 1987, he censured this rejection, first because the interpretation of the Treaty was a matter which concerned only the two Parties of the Treaty, the U.S. and the Soviet Union, and second because the Europeans had not seen the classified negotiating record. As regards the first point, I should like to submit that, as the testing of defensive weapons in space inevitably will have a profound effect on the strategic balance in the world, it is difficult to deny the right of American Allies to express their opinion on an interpretation of the Treaty which would permit such testing.

As to the second point, though admittedly the provision laid down in Agreed Statement D, annexed to the Treaty is ambiguous, and though it is true that the Europeans do not know the negotiation record, the testimony given by the authors of the Treaty, including that of the American Chief Negotiator, Mr. Gerard Smith, who declared that a broad interpretation of the Statement would make the Treaty "a dead letter", has sufficiently clarified the intention of the authors.

Moreover, as has rightly been suggested, the very language of Agreed Statement D which begins "In order to ensure fulfilment of the obligation not to deploy ABM systems..." would also prevent the Parties to the Treaty from deploying systems based on other physical principles than those at the time of the conclusion of the Treaty.

c) Apart from the controversies on the testing of "exotic" weapons in space, there is another issue on which the West-European countries have expressed opinions, divergent from those manifested in the U.S., namely the use and testing of ASAT weapons capable of reaching satellites in high orbit.

As for the U.S., it should be noted that the attitude of the American Administration to the development and testing of these weapons, has - in the last few years - given rise to frequent changes.

In view of the great overlap between anti-satellite technology and that of S.D.I., the question of the use of ASAT's is of crucial importance. Since satellites operating at high altitude are essential for communication, arms control information and early-warning of a strategic nuclear attack, the American Administration originally was strongly in favour of

trying to achieve an agreement with the Soviet Union aimed at at protection of these satellites.

Though in April 1984, *President Reagan* declared that the Administration was exploring the possibility of a limited agreement which could protect these satellites, in August 1985 he informed Congress of the intention to conduct its first test of an ASAT against an object in space, stating that such a test was in accordance with the ABM Treaty. However, as Congress considered the consequences of such testing to be ultimately against strategic interest of the U.S., it passed in December 1985, a resolution halting U.S. tests until 30 September 1986 unless the Soviet Union resumed testing of these weapons before that time. In October 1986, Congress and *President Reagan* agreed to extend the ASAT test ban through the end of Fiscal Year 1987.[7]

But in the course of this year, several developments have taken place which indicate that at least some members of the Administration appear to consider that any extension of the ASAT test ban would be detrimental to national security.

In March 1987, the Defense Department unveiled a plan to restructure the Administration's anti-satellite programme with the goal of deployment. *Mr. Weinberger* said that this programme emphasized the Administration's commitments to a serious imbalance between U.S. and Soviet anti-satellite capabilities. The Pentagon hoped to conduct three air-launched anti-satellite weapons tests against space targets after October 1987.

Though at the time of writing the present article no decision on the lifting or continuation of the ban has yet been taken, it is important to refer to a Bill introduced on 10 March 1987 by a number of Senators aimed at a limitation of ASAT's and to a Joint Resolution of the same date by a number of the House of Representatives requiring the U.S. to continue during fiscal year 1988 the existing U.S.-Soviet Union moratorium on testing ASAT weapons so long as the Soviet Union does the same.

Whereas one cannot tell whether, or at what time, the bills may be passed, in view of the strong support of Congress to the continuation of the present moratorium, there are reasons to expect that an agreement on such continuation could be reached.

Insofar as the Soviet Union is concerned, the concrete deal which the Soviets want most is a continued moratorium on the testing of ASAT's. If the view of American space experts that the Soviets are ahead of the U.S. in ASAT's, having for several years the world's only operational systems, is correct, their insistence on the need of a moratorium on ASAT testing is not surprising.

In the context, it is important to note that during discussions of the Directors of the New York Council of Foreign Relations in Moscow in April 1987, Soviet officials argued that a defensive system would not be

7. For a more detailed exposition of the American and Soviet position on this issue *see* Goedhuis, *The Importance of Preserving and Strengthening the ABM Treaty of 1972*, INTERNATIONAL RELATIONS 475-493 (1986).

vulnerable to Soviet attack. Non-nuclear anti-satellite weapons could knock out elements of the S.D.I. system.

Turning now to the attitude of West European countries regarding the use of ASAT weapons, several of them, including France, Britian, West Germany, the Netherlands and Belgium declared their strong support to the reaching of an agreement by which a ban would be imposed on weapons capable of reaching satellites in high orbit.

As to France, the French representative to the Geneva Disarmament Conference, in June 1984, put forward a proposal along these lines and in November of that year, *Mr. Trudeau,* the former Prime Minister of Canada, called on NATO to support the French plan, also supported by Canada, to ban this kind of ASAT weapons.

It should be noted however that up till now the efforts to arrive at a multilateral agreement on such a ban have not obtained any results.

So far as the legal position of the use of ASAT's is concerned, first it should be noted that in the ABM Treaty no consideration has been given to the use of these weapons. However, there is one provision in this Treaty which is also relevant to such use, namely article V pertaining to limits on space-based and other mobile systems and components.

Another provision which limits the use of ASAT's, has been laid down in article IV of the Outer Space Treaty of 1967, in which the Parties to the Treaty have undertaken not to place in orbit around the Earth any objects carrying nuclear weapons or any other kind of weapons of mass destruction. Both Space Powers have spent considerable amounts of money in the research of high-energy lasers, including X-ray lasers. The use of such type of lasers on ASAT weapons, which may derive their energy from small nuclear explosives, would violate this article. Moreover, the testing of such weapons would jeopardize the 1963 Partial Test Ban Treaty which bans, inter alia, nuclear weapons in outer space.

Another question which arises in this context relates to the permissibility or non-permissibility of stationing ASAT's on the Moon. article III of the Moon Treaty of 1979, reiterating article IV (2) of the Outer Space Treaty of 1967, provides that the Moon shall be used by all Parties exclusively for peaceful purposes. It should be mentioned that up till now, neither of the two main Space Powers have ratified this Treaty.

Though a survey of the fundamental controversies on the meaning of this term, on which an abundant literature exists, would far exceed the scope of this article, suffice it to say that, whereas the U.S. continues to interpret the term "peaceful" as "non-aggressive", the Soviet Union and most West European countries consider the term to mean "non-military". An acceptance of the American interpretation would mean that a stationing of ASAT's on the Moon for defensive purposes would not be illegal. It is submitted however that, on accepting such a position, one would necessarily come into conflict with the provisions laid down both in the Space and Moon Treaty, that the establishment of military bases, installations and fortifications shall be forbidden.

Although great caution is required when one wishes to speculate on the possibility of the U.S. and the Soviet Union arriving at an agreement

on the interpretation of the term "peaceful", it is difficult to imagine that in the short term such an agreement can be reached. Consequently it seems unlikely that the original aim of the authors of the Treaty to arrive at a complete demilitarization of the Moon can be achieved.

d) I come now to another problem on the use of defensive weapons in space, namely the potential military uses on space-stations. On this issue also, strong divergencies exist between the West-European countries and the American Administration.

First a few remarks on the present state of development of these stations will be made. Although there are at this stage several factors which make it difficult to obtain a clear picture of how and when the aim of constructing viable space stations can be achieved, on both sides of the Atlantic, a firm determination exists to vigorously proceed with their development.

As to the United States, *President Reagan*, in January 1984, announced the American aim to establish the first manned space-station in the next decade. Whereas, until the disaster of the Challenger space shuttle in January 1986 and the following accidents with the Titan and Delta rockets, priority has been given to the development of the space shuttle as the sole means of transportation for the aim of establishing these stations, the recognition has been growing that the policy of relying on the shuttle fleet alone as a means of such transportation must be abandoned. Apart from the use of expendable rockets, great pressure is being exerted on the development of an aerospace plane which is believed ultimately to be available for at least some of the tasks which up until now have been carried out by the shuttle. On the pursuance of this project, both in the United States and in Europe, I will make some comments at a later stage.

Before considering the role of the American Allies in the development of the station, I draw your attention to the remarkable progress of the Soviet space-station programme.

The first low-orbiting space-station, called Salyut I was launched by the Soviets in April 1971. Since that time several other Salyut spacecraft were launched, but a new larger orbital station, named MIR (Peace), was launched in February 1986, constituting another important step towards the aim of a permanently manned complex in space. In view of this progress, the Soviets, at this stage, appear to have an important lead in the establishment of such stations.

Now, as regard the role of the West European countries in the development of space stations, in August 1973, a "Memorandum of Understanding" between NASA and a number of European countries was signed for a cooperative programme concerning a Space Laboratory (SKYLAB) in

conjunction with the American space shuttle.[8] The European nations participating in SKYLAB are: West Germany, France, Italy, the Netherlands, Denmark, Austria and Switzerland. It should be noted, however, that as SKYLAB is not a facility that can be used over a multi-year period, it cannot be considered as a true space station.[9]

Two years later, in 1975, a European Space Agency (ESA) was established by merging the European Organization for the Development and Construction of Space Vehicle Launches (ELDO). The Convention regarding the establishment provides *i.a.* that ESA will be used for exclusively peaceful purposes to promote European space cooperation in scientific programs and space application systems. It also provides that the Agency may cooperate with other international organizations and other governments. As to such cooperation, in April 1985, the Council of ESA decided to join the U.S. in design studies for a 12 billion dollar space-station planned for the mid 1990's. It was further agreed to authorize the spending of about 60 million on studies for a space laboratory unit, named "Columbus", which could be plugged into the U.S. core of the projected space station.

ESA countries, while recognizing the importance of cooperating with NASA in the international space station programme, decided at the Rome Space Station Conference in January 1985, to evolve a long term autonomous space capability. Consequently ESA wanted to maintain the option of detaching the module from the space station for autonomous operations. On this issue, a dispute arose between ESA and NASA which, at the time of writing, has as yet not been resolved.

At present ESA is completing final studies of a new multi-million dollar program encompassing space transportation, space station and telecommunications projects that will set the course for European space activities for the next ten to twenty years. This programme, to be presented this year for approval by its members, covers authorization for developments of the Columbus space station, the Ariane launcher and the Hermes space plane.

Insofar as the development of a space plane as a means of transportation to and from space stations is concerned, it should be noted that there are several projects to develop such a plane. As to the U.S., President Reagan referred to the project of a National Aerospace Plane in his February 1986 "State of the Union Address".

In Europe, in December 1986, ten members of the European Space Agency (ESA) agreed to participate in a 35 million dollar development effort for the Hermes space plane, which is expected to enter service in

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8. *See* Galloway *The Relevance of General Multilateral Space Conventions to Space Stations*, SPACE STATIONS 51 (Bockstiegel ed., Carl Heymanns Verlag 1985) .

9. On the SPACELAB, Europe's contribution to the U.S., *see* Bourely, *Agreements between States and with International Organizations*, SPACE STATIONS 71.

the mid 1990's. Britain, having its own project of a Horizontal Take-off and Landing Plane (HOTOL) has as yet not formally joined in the Hermes project, but, as it has realized that HOTOL cannot be financed by Britain alone, it appears to be likely that the British Government will also join in the multi-national project, possibly in the near future.

The Soviet Union, for its part, has a programme for a space plane which is expected to begin full-scale manned flight testing soon.[10]

The crucial and ardently debated question to which attention has to be given now, concerns the potential military uses of space stations. I may start by saying that, in my personal view, it is difficult to imagine that *all* military uses on these stations can be prevented.

So far as the present position of the American Administration is concerned, whereas in the beginning of 1984 the Department of Defense expressed the view that there were no defense requirements for such a station, at a later stage the Department appeared to be convinced that, apart from functions which have both military and non-military objectives, such as remote sensing reconnaissance, communication, weather observation and others, purely military functions on these stations, including ballistic defense operations, cannot be avoided. It can be assumed that the Department of Defense, in taking this position, was also influenced by Soviet manned military demonstrations on the Salyut 7 space station as monitored by U.S. intelligence agencies.

However, the intention of the Department to draw up formal plans for such uses raised considerable concern, not only of the American European Allies, but also of Japan and Canada. They considered that such plans would seriously threaten the international cooperation on the multi-billion dollar project. In this context it should be mentioned that preliminary agreements between the United States and its partners in the development of these stations said that they would be used only for peaceful purposes.

In the beginning of February 1987 the Administration, reacting to this concern, testified that there would be no major weapons systems on the stations but that some Pentagon research could be performed, such as research of semi-conductors. However, two months later, a letter of the Defense Secretary, *Mr. Weinberger,* to Secretary of State, *Mr. Shultz,* revealed high-level disagreement among Administration officials about European, Canadian, and Japanese participation in the station. In this letter, *Mr. Weinberger* counseled *Mr. Shultz* to reserve a place on this station for the military and suggested that if such decision would result in the international partners dropping out of the station programme, the U.S. should be prepared to go forward alone if the price of cooperation would be too high.

When one considers now the Soviet attitude towards the potential military uses of space stations, after President Reagan's announcement on 25 January 1984 that the U.S. would establish its first manned space

10. *See* Aviation Week and Space Technology, 30 March 1987.

station, the Soviet News Agency Tass said in an article published the following day, that the U.S. space station would become a tool of the military. The Soviet assertion, made on several occasions, that its space station would only be used for non-military purposes, has been challenged in several American publications. In an article published in the American journal "Aviation Week and Space Technology" it was postulated that the Soviets intended to provide a self-defense capability for their space stations which would include direct-energy weapons,[11] and in an article published in "Newsweek",[12] it was stated that U.S. intelligence officials had been experimenting with a laser programme similar to the one planned by the U.S.

Attention may also be drawn to the 1985 edition of the U.S. Department of Defense publication "Soviet Military Power", according to which the Soviets have made known their plans to replace Salyut with larger space complexes supporting twenty or more cosmonauts on a permanent basis. Such complexes would enhance their space-based military support and war capabilities.

However that may be, it is difficult to imagine that the Soviets would not try to match any military uses of space stations which the U.S. would consider to be indispensable for its security.

So far as the approach of the West European countries to the military uses of space stations is concerned, there appears to be as yet no firm consensus on this issue.

In the first place, attention should be drawn to the address which *President Mitterand* gave on 7 February 1984 to the two Houses of Parliament in The Hague, in which he called on Western European nations to begin their own work on orbiting defensive space stations. This proposal was apparently aimed as an edge against the unavailability of a corresponding U.S. system.

In November 1985, the importance of building a European military was also stressed by *Mr. Pichaud,* Chief military space engineer for the French General Delegation for the "Armaments Ballistic Missile Office". He suggested that the platform proposed as a joint European project could be served and fueled by the Hermes spaceplane. On the role of such a plane, some comments will be made below.

It should be noted however that, contrary to the above suggestions, the Convention adopted by the European Space Agency, to which reference was made above, provides that the ESA will be used for exclusively peaceful purposes and this includes European programmes for the development of a space station.

I referred to the importance of the projected space plane as a means of transportation to and from space stations. Also, on the question whether such a plane might be used for military purposes, no agreement

11. Aviation Week and Space Technology, 11 June 1984.

12. Newsweek, 11 March 1985.

appears to exist between the U.S. Administration and the West European countries, particularly in the Hermes project.

In the "Guide to the S.D.I.", the enormous potential military value of space planes has been stressed[13] and, as was mentioned in an article in "Aviation Week and Space Technology" of 27 April 1987, the American Department of Defense will fund about three fourths of all the costs.

The European participants in the Hermes project, on the other hand, have outlined its missions solely for science and observation requirements.

Though, as far as I know, the Soviet Union has not made any official statements on its intention to use its space plane for military purposes, it has been suggested that this plane, which is expected to begin full-scale testing soon, will become the world's first space fighter, will provide space station defense, anti-satellite operations, reconnaissance and emergency repair of Soviet satellites.[14]

Attention should be drawn to a remarkable development which happened at the end of December 1986. It was reported that France had contacted the Soviet Union to discuss the technical aspects necessary to make the Hermes space plane compatible with the Soviets MIR orbital space station. If this report were correct, the French contact with the Soviet Union on this issue could mean first, that the French do not consider Hermes to serve military purposes and second, that they also believe that such purposes will not be used on the Soviet MIR station.

But what about the statement made by the French military expert, *M. Pichaud,* with reference to the above, that a European military platform could be served and fueled by the Hermes space plane?

In the preceding pages it has been submitted that in the context of the S.D.I. programme, military uses of space stations cannot be avoided. As space planes are an indispensable element in the development of these stations, it seems difficult to believe that efforts to arrive at an agreement aimed at using space planes only for non-military purposes would have a chance of success.

Proceeding now to the legal aspects of the potential military uses of space stations, it should be noted that a difference exists between the legal rules applicable to space stations in free orbit and those relating to stations on the Moon.

As to the first category, consideration should be given to the legality of two kinds of functions on these stations.

Attention has been given above to functions which have both military and non-military objectives, such as remote sensing, reconnaissance, communication and others. There can be no doubt that such activities will also be carried out on space stations and that - as is

13. *See supra* note 10, at 322.

14. *See* Aviation Week and Space Technology, 30 March 1987, p. 29.

generally recognized - operations like reconnaissance play an important role in the achievement of greater strategic stability.

So far as the legal permissibility of these functions is concerned, there are no rules so far adopted which would restrain or prohibit any of them. And among the host of issues which arise in the potential militarization of space stations, the most crucial one relates to the use of defensive weapons on these stations. It is suggested that, in the context of S.D.I., support of ballistic missile operations of space stations cannot be avoided.

When considering the legal position of such operations, it is the A.B.M. Treaty which has to be taken into account. Article V of this Treaty provides that each Party undertakes not to develop, test or deploy A.B.M. systems, which *inter alia* are space based. As research into such systems is not specified in this article, as has been mentioned above, a general recognition exists not only in the U.S. and Europe but also in the Soviet Union that research on defensive weapons cannot be prevented.

In the preceding pages, comments have been made on the continuous serious divergencies on the *kind* of research which may be permitted under the terms of the Treaty, and these divergencies will make themselves felt also on the research of defensive weapons on space stations.

Though the American Administration considers that the testing of exotic space weapons outside the laboratory is allowed, it is submitted that there can be no doubt that under the rules so far adopted, there is one kind of testing which - also on space stations - would be illegal, namely the testing of X-ray lasers using nuclear explosives. Such testing would infringe both the provisions contained in article IV of the Outer Space Treaty of 1967 as much as those laid down in article I of the Test Ban Treaty of 1963.

As to the legal rules applicable to lunar stations, reference was made above to article IV (2) of the Outer Space Treaty and article 3 of the Moon Treaty of 1972 and to the conflicts of the meaning of the provision that the moon and other celestial bodies shall be used by all Parties to the Treaty, exclusively for peaceful purposes.

An acceptance of the American interpretation of the term "peaceful" as "non-aggressive" would mean that a lunar station could be used for all kinds of military purposes, as long as they could not be considered to be aggressive. But, as has been submitted, such a result would certainly run counter to the intention of the authors of the Treaty and would moreover come into conflict with the provision contained in both the articles referred to above, stipulating that the establishment of military bases, installations, and fortifications on the moon and other celestial bodies shall be forbidden.

III. Turning now to the views expressed by West European Governments on the question whether, and if so when, a separate European space defensive system should be established, it should be noted that as yet no firm consensus on this issue has been attained. Until now the discussions

discussions on the need of such a system have been concentrated on the possible development of an anti-tactical missile plan (ATBM).

As was to be expected, the West European position on this issue is strongly being influenced by the progress made on an agreement between the U.S. and the Soviet Union concerning the elimination of intermediate and short-range nuclear missiles in Europe. But before making some comments on this influence, consideration should be given to statements made by Amercia's European Allies on the ATBM system, *before* the prospective of the achievement of a European Missile Deal between the two main Space Powers. In the first place, reference should be made to a declaration made in October 1986 by the West German Defense Minister, *Mr. Worner*, according to which the West German government was strongly in favour of the establishment of an ATBM system. From statements made be French officials, the conclusion can be drawn that the French Government also was firmly convinced of the need of developing such a system.

On the other hand, the views expressed by members of the Netherlands government have shown that the Netherlands appeared to be in doubt as - at least - to the urgency of the need of evolving this system. In December 1987, the Dutch Minister of Defense, *Mr. van Eekelen* and the Dutch Chief of Staff, *General Huyser*, both warned against an overrating of the threat of tactical nuclear weapons. However, *General Huyser* declared to be strongly convinced that, in the long run, a European defense against these weapons would be inevitable.

Statements made by British officials at that time also demonstrated a certain lack of positiveness of the urgency of a development of the system.

When one now considers the European reaction to the prospects of a European Missile Deal, there are strong indications that the West European Governments increasingly recognize the fact that such a deal not only necessitates a fundamental rethink of long-held assumptions on European defense *in general*, but also that such a deal strengthens rather than lessens the urgency of efforts aimed at reaching a consensus on European space defense.

In this context, reference should be made to an article published at the end of June 1987, in which it was reported that after recent exploratory discussions between American and European defense officials, the U.S. and West European Allies had come closer to an agreement on a collaborative anti-tactical missile programme.[15]

In this article it was further reported that a senior American official of the Defense Department has said that an agreement on specific cooperative ATBM systems could be sorted out within the next six months. I might submit that, when one takes account of the many divergencies between the present American Adminstration and West European countries, in particular those relating to the development of the *kind of*

15. *See* Aviation Week and Space Technology, 29 June 1987, pp. 18-20.

defensive space weapons which are legally permissible, one may have strong doubts that the optimistic prognostications referred to above might come into operation. The obstacles must be overcome before a multilateral agreement expectation can be resolved within the short period of six months.

IV. On the basis of the preceding observations, some conclusions may be drawn.

Consideration was given to the dissimilarities in the attitude of the American Administration and West European countries on three main issues: the first regarding the legal permissibility of the development of space weapons in *general*, the second relating to the development and use of ASAT weapons, and the third pertaining to potential military uses of space stations.

As the present gap between the American Administration and Western Europe on these issues obviously has an unfavorable effect on reaching a collaborative agreement between the U.S. and its Allies aimed at a European space defensive system, some speculations may be made on whether and how this gap might be closed.

As to the first issue, reference was made to the continuing controversies between the present American Adminstration and Congress on the interpretation of the ABM Treaty of 1972. It was suggested that it would seem highly unlikely that, during the remaining months of the present Administration this confrontation could be solved.

On the basis of several statements made by members of the Democratic majority of Congress, however, one might be inclined to assume that a new Administration, instituted after the election at the end of 1988, would not follow the present Administration's broad interpretation of this Treaty.

From these statements, the conclusion can be drawn that while the great majority of the Democratic members of Congress appear to be convinced of the need to continue with the research of space weapons, they would reject the view that under the terms of the ABM Treaty, the testing of these weapons outside the laboratory is legally allowed.

If this assumption would prove to be correct, there would be grounds to believe that, since the American West European Allies also strongly oppose the broad interpretation of this Treaty, a closing of the present gap on this issue might be achieved.

With regard to the second issue, relating to the use of ASAT weapons, attention was drawn to the divergent views expressed by members of the present American Administration on the question whether the development and testing of these weapons would be essential to protect American strategic interests.

In this context, mention was made to the Resolutions passed by Congress in December 1985 and in September 1986, by which a moratorium on U.S. testing of these weapons was accepted. It was suggested that, though at the time of writing the present article, no firm prediction of the continuation of such moratorium could be made, the two

Bills introduced in March 1987 by both members of the Senate and those of the House of Representatives, demonstrated a strong support to such continuation.

If these Bills were accepted, and since the Soviet Union on several occasions has declared their strong attachment to a ban on ASAT's, at least some progress on a bilateral treaty between the main Space Powers on limitations on the testing, deployment and use of these weapons (a Treaty which was proposed in the Bill by the members of the House of Representatives) might be expected.

The enquiry of the attitude of the West European Government towards ASAT weapons, demonstrated that they were strongly in favour of a ban or of, at least, a limitation of these weapons. However, as the use of these weapons have wide implications, not only for the main Space Powers but also for all other countries in the world, the European proposals were aimed at the concluding of a treaty on this issue on a *multilateral basis*.

Though the negotiations between the U.S. and the Soviet Union aimed at the conclusion of a bilateral agreement on the use of ASAT's should be considered to be of crucial importance, the discussions initiated by the Geneva Conference on Disarmament contemplating a possible *multilateral* treaty, should be pursued in parallel to those of the two main Space Powers.

Whereas a cautious optimism might be expressed that on this issue the American and West European attitudes might eventually coincide, it is on the third issue, the potential military uses of space stations, that the greatest obstacles on the achievement of a common approach present themselves.

So far as the present attitude of the U.S. is concerned, it has been mentioned that, though originally the American Department of Defense believed that there were no defense requirements for these stations, at a later stage the Department became convinced that apart from functions on these stations which have both military and non-military objectives, purely military functions, including ballistic defense operations, cannot be avoided. It was suggested that this position was also influenced by Soviet manned military demonstrations on their space stations, which apparently had been monitored by American intelligence agencies.

West European countries, however, have voiced their strong opposition to the use of space stations for military purposes. In the Convention adopted in 1975 by the members of the European Space Agency, it was provided that the Agency would be used for exclusively peaceful purposes and this includes European programmes of a space station.

When the intention of the American Department of Defense became known, considerable concern was expressed by American Allies. They considered that plans on the militarization of space stations would seriously threaten the project aimed at international cooperation in the development of these stations.

But can one expect that the European desired object of limiting the use of space stations exclusively for non-military purposes can be

fulfilled? It was submitted that - as space stations are bound to play an essential role, in particular, in ballistic defense operations - military uses of these stations can hardly be prevented.

A demilitarization of space stations could only be achieved if an agreement would be reached on the prohibition of the development of *all* weapons in space and efforts to reach such an agreement have failed and are bound to fail, at least in any foreseeable future.[16]

Finally, reverting to the remarks made in the preceding pages on the West European position towards the development of an anti-tactical missile system for Europe, it was suggested that though the prospects of a Euromissile deal has resulted in a growing realization of the need not to delay discussions on this development, divergencies continue to exist between the West Europeans on the *urgency* of establishing ATBM systems. When one takes account not only of those divergencies but also of the conflicts between the present American Adminstration and West European countries regarding the *kind* of developments of space weapons in general, which, under the terms of the ABM Treaty, is allowed, an *early* agreement on the establishment of a collaborative anti-tactical missile system can hardly be expected.

Nevertheless, the importance of the exploratory discussion between Amercian and European defense officials on this issue, to which reference has been made, should not be underrated. By studying the options which are available, by dispersing the present confusions and illusions, a clearer perception could be achieved of whether and when an ATBM system should be established.

16. In the address of the present writer, given to the Parliamentary and Scientific Committee of both Houses of the British Parliament in June 1984, it was argued that military competition in space is as inevitable as military competition on earth, sea and in the air. *See* 42 Science in Parliament, No. 179 (October 1984).